



1.4 Benign Oral and Odontogenic Disorders

◆ Key Features

- Pathology may arise from any of the tissue types within the oral cavity—mucosal, glandular, nervous, vascular, immunologic, osseous, and dental.
- Lesions may present symptomatically or be noted on routine screening.
- Squamous papilloma is the most common oral mucosal papillary lesion (2.5% of all oral lesions).
- Benign lesions should be differentiated from malignancy, which when indicated requires biopsy.

Benign oral lesions are common and may be manifestations of local or systemic disorders. The appearance of the lesion, associated symptoms, temporal relationship to other illnesses, and, when necessary, biopsy will enable the appropriate therapy to be determined. Differentiation of benign and malignant lesions is important for proper and expedient care.

◆ Etiology

Lesions may be infectious, inflammatory, neoplastic, traumatic, vascular, or congenital/developmental. They may be isolated phenomena or be related to underlying medical disease or systemic disorder. The pathophysiology varies widely and is mentioned in connection to specific lesions in the Clinical section, below.

◆ Clinical

Signs and Symptoms

Lesions of the oral cavity may present asymptotically or may be noted by the patient or dental provider. Associated symptoms may include pain, bleeding, recurrent trauma, dental pain, malocclusion, or loose dentition. Benign lesions of varied etiology may share morphologic features. Many lesions appear as white plaques (i.e., leukoplakia) or ulcerations. Epithelial

changes involving alteration in maturation (e.g., hyperplasia, dysplasia, carcinoma) often appear white in the moist environment of the oral cavity. Time course of the lesion and associated symptoms may guide the clinician as to the nature of the pathology.

Differential Diagnosis

For patients with oral cavity lesions or periodontal findings, several considerations must be taken into account in generating the differential diagnosis: color of the lesion (e.g., white, red, normal), thickness (e.g., exophytic, plaque, erosive), presence of pain, subsite within the oral cavity/oropharynx (e.g., hard palate, soft palate, gingiva, buccal mucosa, lips), type of apparent tissue involved (e.g., mucosa alone, bony, salivary), and underlying induration or fluctuance. Lesions may involve local factors or may be related to systemic illnesses. Systemic lupus erythematosus, sarcoidosis, Sjögren's disease, Kawasaki's disease, drug reaction, amyloidosis, viral infections, immune deficiencies, Crohn's disease, HIV, and nutritional and metabolic deficiencies are among the possibilities.

Exophytic Mucosal Lesions

Squamous Papilloma

Squamous papilloma is a common benign epithelial growth caused by human papillomavirus (HPV). The lesion is usually pink to red with typical verrucous appearance. These may be singular or multiple (more often in immunocompromised patients). Excision to include the base of the lesion is typically curative, but lesions can recur from incomplete resection or other cells infected with HPV.

Fibromas

Fibromas are nodular swellings with fibrotic submucosal reaction related to trauma or chronic irritation. These may occur anywhere but are more common on the buccal, labial, and lingual mucosa. Excision may be carried out for diagnosis or if repeated trauma is occurring because of the lesion size.

Congenital Epulis

Congenital epulis is a benign lesion occurring along the alveolar region in infants, with a female predominance. Simple excision may be done for comfort or diagnosis.

Lobular Capillary Hemangioma (Pyogenic Granuloma)

Lobular capillary hemangioma (pyogenic granuloma) was long thought to be an inflammatory lesion but has been reclassified as a benign vascular neoplasm. It may be found on any mucosal surface in all demographics, but there is some association with pregnancy. Although vascular in nature, it may become increasingly fibrotic with time. Simple excision is the treatment of choice, but recurrence is not unusual.

Granular Cell Tumor

Granular cell tumor is a benign neoplasm originating with Schwann cells. They most often form on the lingual dorsum and may have some mild surrounding induration. Simple excision is typically curative.

Oral Hairy Leukoplakia

Oral hairy leukoplakia occurs on the tongue, often along the lateral border, and appears as rough white plaques. It is caused by Epstein-Barr virus (EBV) and occurs most often in immunocompromised patients, such as HIV-positive individuals. The appearance can fluctuate. Typically this is asymptomatic, but occasionally some mild discomfort or taste aberration may be present. No specific treatment is required.

Hairy Tongue

In contrast to the preceding, hairy tongue occurs along the tongue dorsum and may be darkly pigmented, pink, or even colored by other agents, such as coffee or mouthwash. It is associated with smoking, poor oral hygiene, local radiation, and antibiotic use. The filiform papillae of the tongue fail to desquamate normally and hence become elongated (or “hairy”). The condition is usually asymptomatic, but some patients relate a tickling sensation or find it unsightly. Treatment involves limiting the potential offending agents, along with tongue brushing or use of a commercially available tongue scraper.

Mucocoeles

Mucocoeles are cystic, ballottable swellings related to salivary gland trauma, often occurring on the lips or buccal mucosa. They may be clear, pink or even bluish in color. They may spontaneously resolve or may be excised to avoid repeated trauma or for diagnosis.

Gingival Hyperplasia

Gingival hyperplasia involves excessive growth of the gingival (gum) tissue. This may occur in response to inflammatory conditions, but when more pronounced, it is more often related to side effects of medications, including phenytoin, cyclosporin, and some calcium channel blockers.

Mesenchymal Tumors

Mesenchymal tumors may arise in the oral cavity. These include hemangioma, leiomyoma, rhabdomyoma, schwannoma, and neurofibroma. Local excision is usually adequate in these lesions, except hemangioma which, in pediatric patients will often regress.

Salivary Gland Adenomas

Salivary gland adenomas may occur in the mucosa of the oral cavity. Pleomorphic and monomorphic lesions may occur. These should be distinguished from minor salivary gland malignant lesions, which are more common.

Surface Mucosal Lesions

Leukoplakia

“Leukoplakia” is a term that may be used to describe any white plaque in the oral mucosa. It typically is related to some epithelial maturation issue, ranging from hyperkeratosis to dysplasia to carcinoma. Leukoplakias may occur as a result of chronic irritation, such as trauma (e.g., dentures, chewing on edentulous alveolus) or tobacco use (e.g., chew/“dip,” smoking). Excisional or incisional biopsy may be indicated for diagnostic purposes to rule out early carcinoma.

Nicotine Stomatitis

Nicotine stomatitis is a leukoplakic lesion, located along the hard and soft palate and caused by the heat from cigar, pipe, or, less commonly, cigarette smoking. Often it is diffuse in nature with white irregular patches mixed with red spots. If consistent historically, it does not require biopsy unless clinical concerns are present.

Fordyce Granules

Fordyce granules are flat to slightly raised yellow clustered lesions occurring on the buccal mucosa and lip. They represent sebaceous glands within the mucosa and require no treatment beyond patient reassurance.

Lichen Planus

Lichen planus is a dermatologic condition that may also involve the oral mucosa. Classically it appears as pink to purple patches with surrounding white “lacy” lines. Often this is asymptomatic, but it may be painful in the erosive subtype. Biopsy is necessary for definitive diagnosis. Treatment is not often necessary for the asymptomatic lesions, but topical or systemic steroids may be necessary in the painful erosive circumstances. A slightly elevated risk of oral squamous cell carcinoma is associated with lichen planus, particularly the erosive subtype. Lichen planus should be monitored during regular semiannual dental visits.

Kawasaki's Disease

Kawasaki's disease is a febrile vasculitis disorder affecting young children. Symptoms include prolonged fever, nonexudative conjunctivitis, uveitis, perianal erythema, “strawberry tongue” with fissures, lymphadenopathy, erythema and edema of the palms and soles, and possible coronary arteritis with possible aneurysm formation. This last issue should be evaluated with echocardiogram, as it can lead to death. The etiology remains unknown, but some suspect an infectious trigger. Treatment includes supportive care, monitoring for cardiac issues, and intravenous immunoglobulins.

Tattoo

Tattoo may be present on the mucosa from embedded pigmented materials or dental amalgam. These may be located away from the point of introduction secondary to uptake of the material and migration by macrophages. No treatment is necessary.

Racial Pigmentation

Racial pigmentation of the oral mucosa involves diffuse and symmetric melanotic coloration, most commonly on the gingival and labial regions. This should be contrasted to melanosis, as follows.

Melanosis

Melanosis is increased melanin production from irritants (e.g., smoker's melanosis), genetic disorders (e.g., Peutz-Jeghers's syndrome), endocrine disorders (e.g., Addison's disease), or true neoplasm (e.g., melanotic nevus, melanoma).

Ecchymosis, Hematoma, and Petechiae

Ecchymosis, hematoma, and petechiae are different types of collections of extravascular blood. Ecchymoses are bruises and may be purple, red, or blue. Hematoma is a collection of blood in the soft tissues, with swelling and a purple to dark blue color. Petechiae are round and small red lesions. Each of these will typically resolve without treatment. However, unless a specific etiology is known (e.g., local trauma), concern for coagulopathy should be investigated. An expanding hematoma of the floor of the mouth or pharynx can potentially cause airway compromise. Therefore, these patients should be monitored for worsening or be prophylactically intubated to prevent airway compromise.

Candidiasis

Candidiasis is a local fungal infection and may be asymptomatic or painful. The lesions typically appear as multiple white plaques on an erythematous base, with the white material able to be scraped off. Predisposing factors include prior radiation to the head and neck, xerostomia, antibiotic therapy, dentures, steroid inhaler use, and immunosuppression (e.g., diabetes, AIDS). Treatment often involves topical antifungals, although systemic therapy may be useful in particular cases.

Erosive Mucosal Lesions

Mucositis

Mucositis is a general term for inflammation of the oral epithelium. This may be caused by medications, radiation, chemotherapy, and local irritants. The mucositis may be generalized or patchy. Mucosal erythema or erosion may be present, but it is typically superficial. Deeper ulcerations are concerning for alternative diagnoses.

Viral Stomatitis

Viral stomatitis represents multiple possible infectious agents. Herpes simplex virus may cause vesicular and ulcerative lesions, typically associated with pain. The virus typically lies latent and may cause localized eruptions periodically. Primary herpetic gingivostomatitis occurs in children, and less often in adults. It is associated with tender cervical adenopathy, fever, and malaise. Antiviral agents may be helpful in symptomatic treatment. Varicella primarily affects the skin, but reactivation of varicella (herpes zoster) may involve sensor nerve distributions of the skin and oral mucosa. Herpangina caused by coxsackievirus

A involves fever, malaise, pharyngitis, nausea, and diarrhea, with vesicular or ulcerative lesions along the soft palate and tonsillar pillars. Hand, foot, and mouth disease is also caused by coxsackievirus and involves similar symptoms along with rash of the hands and feet. Rubeola is associated with Koplik spots, which appear as red macules with white centers along the buccal mucosa. Rubeola causes more severe systemic illness but is rare, given vaccinations.

Recurrent Aphthous Ulcers (Canker Sore)

Recurrent aphthous ulcers (canker sores) are common ulcerative lesions with a quick onset and typical 7 to 14 day course. They occur on nonkeratinized mucosa, such as the lip, buccal mucosa, unattached gingiva, soft palate, and floor of the mouth. The etiology is unclear, although some family history is not uncommon. Occurrences may be sporadic or related to psychosocial stressors, illness, and menstrual cycle. The lesions are typically painful at first, and the pain may abate before the ulcer is fully healed. They are shallow ulcers covered with a yellow to gray membrane. Treatment may be supportive, although topical steroids speed resolution. Most lesions are singular or few but may occur in a "herpetiform" fashion. "Major" aphthous ulcers are large and can be debilitating. This form may be associated with longer duration and healing with scarring.

Pemphigus Vulgaris

Pemphigus vulgaris is an autoimmune disorder in which the patient develops antibodies to the epithelial desmosomes, creating a loss of cellular adhesion and subsequent sloughing. Bullae form but are often quickly ruptured within the oral cavity. Nikolsky sign may be present on intact bullae. Incisional biopsy with immunofluorescent staining is required for diagnosis. Treatment includes immunosuppressive agents, including corticosteroids.

Pemphigoid

Pemphigoid is similar to pemphigus, but with the antibodies directed at the basement membrane. A Nikolsky sign is less common but may be present. Scarring may occur in this condition.

Benign Migratory Glossitis

Benign migratory glossitis may also be known as geographic tongue. The disorder manifests as lingual erythema, papillary atrophy, fissures, and serpiginous white hyperkeratotic regions. These findings wax and wane over time. The etiology is unknown, but the condition has been associated with psoriasis, granulomatous disease, and certain HLA types. Diagnosis is made through classic features on examination. Many cases are asymptomatic, but some patients have sensitivity to spicy or hot foods. If the condition is severe, possible topical retinoids or corticosteroids have been tried.

Necrotizing Sialometaplasia

Necrotizing sialometaplasia is a process involving salivary acini that undergo ischemia and necrosis. The lesions may involve swelling with some central necrosis and ulceration, and because of this appearance, they can be mistaken for squamous cell carcinoma. Incisional biopsy should be done for diagnosis, but the lesion typically spontaneously resolves.

Vitamin Deficiencies

Certain vitamin deficiencies may have oral manifestations. Deficiencies of vitamin B₂ (riboflavin) and vitamin B₃ (niacin) may be associated with angular cheilitis and with magenta or red-colored tongue with filiform papilla atrophy. Vitamin B₁₂ (cobalamin) deficiency may produce angular cheilitis as well but also aphthous ulcerations, general mucositis, glossitis, and taste aberration.

Odontogenic Lesions

Odontogenic infections may occur in any tooth-bearing region and may extend into the neck.

Periodontal Abscess

Periodontal abscess occurs when infection accumulates in the periodontal space. This may present as swelling and pain around a tooth or tooth remnant. Bony erosion may be present. Spontaneous drainage may occur. Incision and drainage with dental follow-up is recommended.

Periapical Abscess

Periapical abscess is an infection within the periapical region of the tooth, which can be seen on radiography. These may be asymptomatic or may present with surrounding cellulitis, pain, or drainage. If a fistula tract leads to the oral mucosa, a soft tissue swelling may occur, called a parulis or gumboil. An associated soft tissue abscess should be drained, but otherwise therapy includes possible dental extraction or root canal by a dental professional.

Torus Mandibularis and Torus Palatini

Torus mandibularis and torus palatini are common bony protrusions from the mandibular bone and hard palate, respectively. Torus palatini occurs along the midline and may be broad and smooth or may have a nodular appearance, sometimes with a narrower base. Torus mandibularis occurs along the medial mandible from the region above the mylohyoid muscle attachment. No treatment is needed, although they can complicate the fitting of dentures.

Exostoses

Exostoses are benign bony protuberances located on the buccal aspect of the upper (more common) or lower alveolus. The etiology is not clear, although it may involve bruxism and genetic contributors. As with torus mandibularis and torus palatini, no treatment is required, although, again, they may interfere with denture fitting. If larger, they may also contribute to periodontal disease.

Odontogenic Tumors and Cysts

Some odontogenic tumors and cysts may present with oral swelling or distortion of the alveolus, while others may be noted only on imaging studies. On oral examination they are not classically distinguished.

Odontogenic Keratocyst (OKC)

An odontogenic keratocyst (OKC) is a developmental cyst from the dental lamina that occurs in either the upper or lower jaw. It appears as a

well-defined radiolucency in the bone. The cyst is lined with a parakeratinizing epithelium. The OKC can cause local aggressive destruction of bone and has a moderate recurrence risk. Resection can be conservative (e.g., enucleation curettage) or aggressive (e.g., complete resection). Recurrence rates are higher for conservative therapy, although a balance of recurrence versus surgical morbidity must be determined. This lesion may be associated with nevroid–basal cell syndrome (Gorlin's syndrome).

Dentigerous Cyst

A dentigerous cyst is a cystic lesion associated with the crown of an unerupted tooth. The cyst is lined with enamel epithelium. The cyst may become large enough to jeopardize mandible stability. Treatment may include excision of the cyst and origin tooth.

Calcifying Epithelial Odontogenic Tumor (Pindborg's Tumor)

A calcifying epithelial odontogenic tumor (Pindborg's tumor) is a rare odontogenic tumor of unclear tissue origin. It is slow growing and favors the posterior mandible. The radiography shows a heterogeneous mixed osseous pattern. Complete excision is recommended.

Fibrous Dysplasia

Fibrous dysplasia is a dysplastic process of bone growth, with a mixed pattern of osseous and fibrous tissue. Generally, it is self-limited and slow growing. The lesion may cause cosmetic deformity or impingement on other structures, depending on location. Resection is reserved for functional or severe cosmetic concerns.

Ossifying Fibroma

An ossifying fibroma is a slow-growing, well-circumscribed tumor consisting of lamellated bone with fibrous stroma, which can act aggressively locally. It may appear similar to fibrous dysplasia on imaging. Treatment favors resection, given the concern of more aggressive growth.

Ameloblastoma

Ameloblastoma is a benign tumor of ameloblasts (enamel-forming cells) more common in the mandible. It is more common in men in their 30 to 40s. The lesion tends to grow in an expansile fashion. Radiographically it has a "soap bubble" appearance, with multiloculated features. Resorption of tooth roots may be seen. Though histologically benign, these tumors can be aggressive and therefore complete surgical resection is the treatment of choice, as the tumor is poorly sensitive to radiation and chemotherapy.

Systemic Disorders

Amyloidosis

Amyloidosis may occur locally in the oral cavity or as part of a systemic illness. Classically, amyloid (immunoglobulin light chains) deposits in the tongue, buccal, and gingival tissue. This may cause expansion and induration. With severe tongue involvement, this may prevent the tongue fitting

in the oral cavity or being retrodisplaced with airway restriction. Diagnosis is via biopsy, with “apple-green” birefringence on Congo red staining with polarized light microscopy. Treatment may involve systemic therapy, but local resection may assist if airway obstruction is an issue.

Sarcoidosis

Sarcoidosis is a systemic granulomatous disease affecting many organ systems, with predilection for the respiratory system. Salivary glands and oral mucosa may be involved. Sarcoid lesions may appear as red or violaceous nodular swellings of the lips, palate, and gingiva. Xerostomia may also be present secondary to salivary gland involvement and dysfunction. Biopsy is diagnostic and should show noncaseating granulomatous vasculitis. Serum angiotensin-converting enzyme (ACE) levels may be elevated in active disease. Treatment is with corticosteroids and cytotoxic drugs.

Systemic Lupus Erythematosus

Systemic lupus erythematosus is a systemic autoimmune disorder that can involve multiple organs, including ulcerative mucosal lesions.

Sjögren's Syndrome

Sjögren's syndrome is an autoimmune disease affecting the exocrine glands, primarily salivary and lacrimal. This results in keratoconjunctivitis sicca. Primary Sjögren's syndrome occurs in isolation, whereas secondary Sjögren's syndrome is associated with another connective tissue disorder. Diagnosis may be made by minor salivary gland biopsy, usually from the lip, looking for lymphocytic infiltrate. Serology may demonstrate SS-A/Ro and SS-B/La antibodies. Treatment is symptomatic, with an emphasis on good dental care. Five percent of Sjögren's syndrome patients will develop a lymphoid malignancy, most commonly non-Hodgkin's lymphoma.

◆ Evaluation

A detailed history of the lesion and any associated symptoms should be elicited. Past medical history should be taken, focused on possible inflammatory disorders, malignancy, autoimmune or connective tissue disorders, immunodeficiencies, and nutritional or metabolic conditions. The use of tobacco in any form should be noted. Dental and denture history may be helpful in odontogenic disorders and in some traumatic concerns (e.g., poorly fitting dentures, edentulous mastication, fractured teeth causing trauma to buccal mucosa or tongue).

Oral cavity examination should include all mucosal surfaces, keeping in mind that some regions require manipulation to evaluate (e.g., gingivolabial sulcus, posterior floor of mouth). In addition to inspection of the mucosal surfaces, the dentition should be evaluated. Palpation of the floor of mouth; the tongue, including the base; and, when appropriate, the buccal and labial structures should be performed to identify depth of involvement of a lesion. Lesions can be initially characterized by their color (e.g., leukoplakia, erythroplakia), depth (e.g., ulcerative/erosive, plaque, exophytic, expansile), and

location (e.g., gingival, palatal). These features allow refining a differential diagnosis and evaluation plan.

If any question exists as to the nature of the lesion, biopsy is an appropriate evaluation tool. Most biopsies can be sent for routine histopathologic evaluation, but some diagnoses require special techniques and staining. Examples include pemphigus versus pemphigoid, where immunoglobulin staining is done and tissue should be sent fresh.

CT and MRI imaging may be indicated if there is concern for the depth of the lesion, involvement of deeper structures, or proximity to vital structures or if the concern is for submucosal disease presenting luminally. CT provides good soft tissue resolution and the relationship to underlying osseous structures, including abutment to the bone, erosion/invasion of the bone, or even development from the bone itself. While some use of ultrasound intraorally has been described for evaluation of peritonsillar abscess identification and fluid collection, this requires clinician training in the technique. Rarely other imaging such as angiography and sialography may be indicated.

◆ Treatment Options

Individual disease treatments are outlined in the preceding section with the corresponding diseases. The concern of oral cavity squamous cell or minor salivary gland carcinoma must also be considered, and if questions remain after a detail exam and history, biopsy is advised.

1.5 Temporomandibular Joint Disorders

◆ Key Features

- After toothache, temporomandibular joint (TMJ) disorders are the most common cause of facial pain.
- Most patients respond to conservative management.
- Surgical treatments can repair or replace severe damage.

Disease of the temporomandibular joint (TMJ) is often classified as myogenous or arthrogenous, involving a primary problem of muscle tension versus joint anatomy. Patients present with pain, headache, discomfort with chewing, and popping of the TMJ.

◆ Anatomy and Physiology

The TMJ connects the jaw and skull at the condyle of the mandible and the squamous portion of the temporal bone. The articular surface of the

temporal bone is formed by the articular eminence anteriorly and the glenoid fossa posteriorly. The articular surface of the mandible consists of the top of the condyle. The surfaces are separated by an articular disk, the meniscus. The joint meniscus is made of fibrocartilage and is important for smooth joint function. The meniscus has a thick anterior band, a thin intermediate zone, and a thick posterior band; pathologic changes can contribute to TMJ disorders. A dense fibrous capsule surrounds the joint. Joint motion involves both a rotational component and a sliding translational component.

◆ Epidemiology

TMJ disease is more common in women, with a female-to-male ratio of 4:1. The highest incidence is among young adults, aged 20 to 40 years. The disorder is seen more frequently in persons of European descent than in persons of African descent.

◆ Clinical

Signs and Symptoms

Signs and symptoms include TMJ pain, earache, joint popping, joint locking, crepitus, tenderness, and spasm of the mastication muscles. Headache, neck ache, and tooth sensitivity are common. A history of poor sleep or other sleep disorders is common.

Differential Diagnosis

For patients presenting with complaints in the TMJ region, important considerations, in addition to TMJ disorders, include disease involving the ear, such as otitis externa or otitis media, and sources of referred otalgia such as malignancy of the larynx or pharynx. TMJ disorders may be related to rheumatoid arthritis, degenerative joint disease, ankylosis, dislocations, infections, congenital anomalies, and neoplasm. Dental malocclusion, jaw clenching, bruxism, increased pain sensitivity, and stress and anxiety should be considered. TMJ disorders may be mistaken for migraine, or these may be concurrent disorders.

◆ Evaluation

History

As in other areas, a careful history is important to guide the clinical assessment. Particular attention to prior dental problems, dental procedures, and head and neck injuries is important. Prior diagnosis of psychologic disorders, chronic pain, migraine, or other headache disorders should be noted. A detailed, accurate medication list is required, with attention to chronic analgesic or anxiety medication use.

Physical Exam

A full head and neck examination is performed. It is necessary to assess for evidence of active otologic disease, which could be a source of pain. Also, it is important to visualize the pharynx and larynx to exclude an obvious

lesion that could be a source of referred pain. Assess for dental malocclusion, abnormal dental wear, absent teeth, and visible clenching or spasm of the ipsilateral neck muscles. Evaluate jaw motion, which may be reduced in TMJ disease. Normal range of motion for opening is 5 cm. The joint should be palpated, inferior to the zygomatic arch 1 to 2 cm anterior to the tragus, in both open and closed positions. The examiner should feel for muscle spasm, muscle or joint tenderness, crepitance, and limitation of joint motion or asymmetry of motion between sides.

Imaging

CT can provide information on degenerative changes, bony irregularity of the joint, erosions, and fractures. MRI can provide detailed evidence of soft tissue derangements, particularly of the articular cartilage. Imaging is recommended prior to consideration of surgery.

Pathology

Two categories of TMJ disease exist:

1. *Myogenous temporomandibular disease*: Muscular hyperactivity and dysfunction due to dental malocclusion. Psychological factors may be contributory, such as anxiety leading to habitual clenching of the jaw. Factors contributing to muscle spasm include malocclusion, jaw clenching, bruxism, increased pain sensitivity, personality disorders, stress and anxiety, and a history of trauma.
2. *Articular temporomandibular disease*: Joint dysfunction related either to (a) displacement of the meniscus disk or (b) diseases causing degenerative changes to the joint anatomy. Meniscus displacement is the most common cause. Abnormal anterior displacement of the posterior band between the condyle and the eminence leads to signs and symptoms. Muscular hyperactivity/spasm in this situation is secondary. Conditions causing degenerative changes to the joint anatomy include rheumatoid arthritis, degenerative joint disease, ankylosis, dislocations, infections, trauma, congenital anomalies, and neoplasms.

◆ Treatment Options

Medical

Most patients can be managed with conservative treatment, involving joint rest, anti-inflammatories, muscle relaxants, dental occlusal splints/nightguards. Joint rest is achieved with the use of a soft diet and the avoidance of chewing gum. Nightguard splints can be fashioned using dental impressions and will likely reduce nighttime bruxism and masticator muscle clenching.

The main goal of physical therapy is to stabilize the joint and restore mobility, strength, endurance, and function. Modalities include relaxation training, friction massage, and ultrasonic treatment.

Relevant Pharmacology

- Antiinflammatory medication for 4 weeks, then taper:
 - Ibuprofen 200 to 400 mg orally every 4 to 6 hours; not to exceed 3.2 g/d
 - Naproxen 500 mg orally, followed by 250 mg every 6 to 8 hours; not to exceed 1.25 g/d
- Muscle relaxants:
 - Cyclobenzaprine (Flexeril, Ortho-McNeil-Janssen Pharmaceuticals, Titusville, NJ) 20 to 40 mg/d orally divided into two or four doses; not to exceed 60 mg/d
 - Diazepam (Valium, Roche Pharmaceuticals, Nutley, NJ) for mild spasms: 5 to 10 mg PO every 4 to 6 hours as needed
 - Adverse effects include sedation, depression, and addiction.
- Other:
 - OnabotulinumtoxinA (Botox, Allergan, Dublin, Ireland) has been injected to reduce muscle spasm.

Surgical

- Absolute indications: tumor, severe ankylosis
- Relative indications: failed conservative management

Arthrocentesis

Arthrocentesis is a minimally invasive procedure, usually performed under either IV sedation or general anesthesia. A 22-gauge needle is inserted in the superior joint space, and a small amount of saline is injected to distend the joint space, after which the fluid is withdrawn and evaluated. With reinjection, the joint is then lavaged; steroids and/or local anesthetics can be injected into the joint space.

Arthroscopic Surgery

Indications include internal derangements, adhesions, fibrosis, and degenerative disk changes. Arthroscopic lysis and lavage can be a minimally invasive alternative to open procedures.

Arthroplasty

Arthroplasty is open TMJ surgery, including disk repositioning, discectomy, and joint replacement. Disk repositioning is used when the meniscus has become displaced. Under general anesthesia, an incision is made to access the joint, and the displaced disk is repositioned with or without plication. Repair of surrounding ligaments may be required.

Discectomy is indicated for TMJ internal derangement involving disk position and integrity; it is typically done as an open procedure under general anesthesia.

In cases of articular eminence anatomic derangement, articular eminence contouring, a procedure to reduce and smooth the articular eminence, may be indicated.

If TMJ damage cannot be repaired, the patient may be a candidate for removal and replacement of part or all of the TMJ components. Conditions requiring surgery may include severe degenerative disease, congenitally deformed TMJs, and advanced rheumatoid arthritis. Options for a partial joint replacement are (a) articular fossa replacement and (b) mandibular condyle replacement using either autologous bone, such as a rib, or a metal prosthesis. In a total joint procedure, the condyle and fossa are both replaced using prosthetic components.

Complications

A mandible dislocation may occur iatrogenically, typically during intubation, the use of a mouth prop during oral cavity and oropharyngeal surgery, or endoscopy. Typically the condylar head is dislocated anteriorly and sits anterior to the articular eminence. Typically, the jaw can be relocated manually. Support the head. If the patient is awake, give diazepam and consider morphine to provide relaxation and analgesia of muscle spasm so as to permit reduction. Place thumbs behind the last molar on either side of the mandible and grasp the inferior surface of the mandible with fingers on each side. Exert downward pressure on the lower molars to free the condyle from its entrapped position anterior to the articular eminence. Ease the mandible posteriorly to return it to its anatomic position. The teeth should close readily.

◆ Outcome and Follow-Up

In most cases the prognosis is good, and patients respond to conservative management. Surgical outcomes tend to be favorable in properly selected cases.

1.6 Geriatric Otolaryngology

◆ Key Features

- The elderly population is expanding rapidly worldwide.
- The elderly often present with multiple comorbidities.
- Disease presentation in the elderly may be atypical.
- For safe performance of surgery in the elderly, thorough preoperative management is important.

◆ Epidemiology

The proportion of the U.S. population aged 65 and older is predicted to expand from 12.4% in 2000 to 19.6% (or 71 million persons) in 2030. Moreover, the over-80 age group is expected to expand from 9.3 million

Table 1.5 General approach to the elderly patient

History of present illness:

Patient may be a poor historian; obtain primary care records if possible.

Medications:

Obtain accurate current medication list; consider polypharmacy issues.

Social:

DNR status; end-of-life care wishes, advance directives; tobacco and alcohol use; functional status; elder neglect and abuse.

Preoperative:

Careful consideration of risk factors. Testing including CBC, PT, PTT, BMP, chest X-ray, 12-lead ECG. Consider cardiac echo or stress test if risk factors.

Anesthesia:

Consider local, local with IV sedation, versus general.

Examination:

Ancillary testing to consider: audiology, VNG study, videostroboscopy, modified barium swallow.

Abbreviations : DNR, do not resuscitate; CBC, complete blood count; PT, prothrombin time; PTT, partial thromboplastin time; BMP, basic metabolic panel; ECG, electrocardiogram; IV, intravenous; VNG, videonystagmographic.

persons to 19.5 million by 2030. Many diseases affecting the elderly involve otorhinolaryngologic care. Comorbidities such as cardiovascular disease, renal disease, and polypharmacy may increase the complexity of safe and effective medical and surgical care for these patients.

◆ **Clinical**

The clinical approach to the geriatric otolaryngology patient should follow the same general organization as is used for other patients. However, certain areas of the evaluation require specific attention (**Table 1.5**), as discussed in the Evaluation section.

Diseases

Several aspects of otolaryngology—head and neck surgery are directly impacted by aging. Age-related changes may affect hearing, balance, voice, swallowing, nutrition, olfaction, sleep quality, and cosmetic concerns. Many head and neck neoplasms also have an increased incidence with advanced age. A consideration of some common conditions, by area, follows.

Otology

- Ceruminosis
- Serous otitis
- Presbycusis
- Tinnitus
- Balance dysfunction
 - Benign paroxysmal positional vertigo (BPPV)
 - Ménière's disease

- Medication side effects
- Nonvestibular dizziness (related to loss of sensation, proprioception, neuromuscular coordination)

Rhinology

- Presbyosmia
- Rhinitis
- Epistaxis
- Sinusitis
- Nasal obstruction from structural collapse (e.g., ptosis, valve collapse)

Laryngology and Oropharyngeal

- Presbylarynx
- Neurolaryngeal disorders
- Xerostomia
- Dysphagia
- Aspiration
- Reflux

Endocrine

- Hyperparathyroidism with osteopenia
- Thyroid disorders or malignancy

Head and Neck

- Upper aerodigestive tract malignancy
- Parotid neoplasm
- Cutaneous malignancies of the head and neck

◆ **Evaluation**

History

Obtaining a good history from a debilitated, chronically ill elderly patient or a patient suffering from dementia may be difficult. If possible, input from a close relative or friend may be helpful. In addition, the primary care provider (PCP) and/or facility (e.g., nursing home) records should be obtained. For instance, the patient may be unable to state how long a lump has been present, but knowledge of the duration or growth rate of a mass will have a direct impact upon how the mass will or will not be evaluated. Previous medical or surgical treatments for the current problem might be unknown to the patient but are of great relevance. The same is true of previous testing or evaluation, such as laboratories or imaging. The general past medical and surgical history should be thorough; again, PCP records should be sought. It is not uncommon for the elderly patient to forget to mention important facts, such as the presence of a pacemaker, that are crucial to consider when planning procedures such as surgery or an MRI scan.

Medication History

The importance of obtaining an accurate, current medication list cannot be overstated. Medication reconciliation, or ensuring that pretreatment, current treatment, and posttreatment medication lists are accurate, has become an issue of great importance to regulation bodies such as the Joint Commission because of emerging data regarding the consequences and frequency of medication errors. Fortunately, the growing use of electronic medical records is making legible and detailed medication lists more rapidly available. In the elderly, polypharmacy is a widespread problem. Complaints such as dysequilibrium, xerostomia, dysphagia, dyspepsia, and parosmias can be a consequence of medication side effects. This should be considered during the assessment of the elderly. Also, when surgical therapy is being considered, the accurate medication list is critical. The use of cardiac medications such as β -blockers must be considered perioperatively in accordance with current (rapidly evolving) guidelines. Medications such as warfarin, aspirin, rivaroxaban, or clopidogrel must be considered regarding potential risks associated with holding medications preoperatively. Herbs, nutritional supplements, and any complementary and alternative medicine treatments should not be overlooked.

Social History

A key factor to consider when treating elderly patients involves a consideration of end-of-life care wishes. Often the otolaryngologist is not directly involved in these time-consuming and emotional conversations, which may involve multiple family members, close friends, and clergy. Ideally, the PCP can and should address these issues with clearly written documentation. When considering surgical intervention in the elderly, one should be sure that advanced directives have been discussed and properly documented.

Physical Exam

The head and neck exam in the elderly patient does not, in general, differ from that in the general population. There are some generalizations to consider. The incidence of presbycusis or other forms of hearing loss is such that most geriatric patients should be considered for screening audiologic assessment. The incidence of hyposmia in the geriatric group, especially the over-80 group, is exceedingly high. The use of videostroboscopy in the geriatric patient with voice complaints can be of great value in elucidating stiffness and atrophic or other subtle vocal fold changes, thus facilitating appropriate voice rehabilitation choices. Although the otolaryngologist does not generally focus on the physical exam outside of the head and neck, the ability to examine briefly for obvious signs of congestive heart failure, such as peripheral edema or jugular venous distention (JVD), may prompt appropriate preoperative testing.

Review of Systems

Review of systems is especially important in stratifying preoperative risk. Angina, dyspnea on exertion, chronic renal disease, liver failure, and

Table 1.6 Conditions used to identify patients at risk for increased perioperative morbidity and mortality

Cardiovascular: Coronary artery disease, congestive heart failure, presence of arrhythmias, peripheral vascular disease, severe hypertension, recent MI
Respiratory: Smoking history >20 pack-years, morbid obesity, preexisting pulmonary disease, PO ₂ <60 or PCO ₂ >50
Renal: Renal insufficiency (BUN >50, creatinine >3.0)
Hepatic: Cirrhosis, hepatitis
Endocrine: Diabetes mellitus, hyperthyroidism or hypothyroidism, chronic steroid therapy
Hematologic: Coagulopathy, anemia, thrombocytopenia
General: Age >70, bedridden/debilitated, malnutrition, emergency surgery

Abbreviations: MI, myocardial infarction; BUN, blood urea nitrogen.

severe chronic obstructive pulmonary disease (COPD) all must be factored in when considering surgery. General surgeons traditionally have used classifications such as the Goldman criteria to consider major surgical risk factors (**Table 1.6**). An important factor to assess in the elderly is the frailty factor. Frailty is the decline in the cognitive and physical state with reduced physiologic reserve over multiple organ systems. It is not a disease or syndrome but rather a variable state of illness and functional status. Frailty can be a strong predictor of adverse postoperative morbidity and mortality, as well as a predictor for the need of long-term institutionalization. Multiple frailty indexes have been proposed, with most focusing on factors such as fatigue, weight loss, comorbid illnesses, and restrictions on normal ambulation.

It has been stated that “there is nothing like an operation or an injury to bring a patient up to chronological age.” The elderly patient with potentially limited reserve, impaired organ function, and/or polypharmacy requires particular attention, especially when considering surgical intervention. The reader is directed to resources from the recently formed American Society of Geriatric Otolaryngology (ASGO); for mission, meeting, membership, and other information about ASGO, go to www.geriatricotolaryngology.com. Also, consult the online e-book from the American Academy of Otolaryngology–Head and Neck Surgery Foundation, *Geriatric Care Otolaryngology*, at www.entnet.org. As the aging population expands, and economic and physician workforce issues evolve, there is likely to be increasing need for practicing otolaryngologists to provide appropriate and efficient geriatric care.